

7-mavzu: To'la differensial tenglamalar

Reja:

1. To'la differensial tenglama

2. To'la differensial tenglamaga keltiriladigan tenglamalar

1^o. To'liq differensialli tenglama. Agar ushbu

$$M(x, y)dx + N(x, y)dy = 0 \quad (1)$$

differensial tenglamaning o'ng tomoni biror $u(x, y)$ funksiyaning to'la differensialidan iborat bo'lsa, ya'ni

$$Mdx + Ndy \equiv u \equiv \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

ko'rinishda bo'lsa, (1) tenglama *to'liq differensialli tenglama* deyiladi.

Teorema. (1) tenglama to'liq differensiallanuvchi tenglama bo'lishi uchun x va y o'zgaruvchining biror bir bog'lamli D sohada

$$\frac{\partial M}{\partial y} \equiv \frac{\partial N}{\partial x} \quad (2)$$

shartni bajarilishi zarur va etarli.

(1) tenglamaning umumiy integrali $u(x, y) = C$ ko'rinishga ega yoki

$$\int_{x_0}^x M(x, y)dx + \int_{y_0}^y N(x, y)dy = C.$$

Misol 1. Differensial tenglamani yeching:

$$(\sin xy + xy \cos xy)dx + x^2 \cos xy dy = 0.$$

Yechish. Berilgan tenglama to'liq differensialli tenglama ekanligini tershiramiz:

$$\frac{\partial M}{\partial x} = \frac{\partial}{\partial x}(\sin xy + xy \cos xy) =$$

$$= x \cos xy + x \cos xy - x^2 y \sin xy = 2x \cos xy - x^2 y \sin xy,$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(x^2 \cos xy) = 2x \cos xy - x^2 y \sin xy.$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Demak, shart bajariladi. Shunday qilib, berilgan tenglama to'liq differensialli tenglama va

$$M = \frac{\partial u}{\partial x} = \sin xy + xy \cos xy, \quad N = \frac{\partial u}{\partial y} = x^2 \cos xy$$

teng. Shuning uchun

$$u(x, y) = \int (\sin xy + xy \cos xy)dx + \varphi(y).$$

Bu yyerda $\varphi(y)$ hozircha noma'lum funksiya.

Integrallab

$$u(x, y) = x \sin xy + \varphi(y)$$

topamiz. Topilgan $u(x, y)$ funksiyaning $\frac{\partial u}{\partial y}$ xususiyl hosilasi $x^2 \cos xy$ teng bo'lishi kerak. Shuning uchun

$$x^2 \cos xy + \varphi'(y) = x^2 \cos xy$$

munosabat o'rinli bo'lishi uchun $\varphi'(x) = 0$ bo'lishi kerak. Bundan esa $\varphi(x) = C$ ekanligi kelib chiqadi. Demak,

$$u(x, y) = x \sin xy + C.$$

Berilgan differensial tenglama

$$x \sin xy = C$$

umumiy integralga ega.

Ba'zi differensial tenglamalarni integrallashda hadlarni shunday guruhlash mumkinki, osongina integrallanadigan kombinatsiyalarni olish mumkin.

Misol 2. Differensial tenglamani yeching:

$$(x^3 + xy^2)dx + (x^2y + y^3)dy = 0. \quad (4)$$

Yechish. $\frac{\partial M}{\partial x} = 2xy, \frac{\partial N}{\partial y} = 2xy$ (2) mubosabat bajarilganligi uchun berilgan tenglama

to'liq differentsilli tenglama bo'ladi. Bu tenglamani guruhlash yo'li bilan osongina $du = 0$ ko'rinishga keiltirish mumkin. Shu maqsadda bu tenglamani quyidagi ko'rinishda yozib ilamiz:

$$x^3 dx + xy(ydx + xdy) + y^3 dy = 0.$$

Differensial qoidalariga ko'ra

$$x^3 dx = d\left(\frac{x^4}{4}\right), xy(ydx + xdy) = xy d(xy) = d\left(\frac{(xy)^2}{2}\right), y^3 dy = d\left(\frac{y^4}{4}\right)$$

munosabatlar o'rinli. Shuning uchun (4) tenglamani

$$d\left(\frac{x^4}{4}\right) + d\left(\frac{(xy)^2}{2}\right) + d\left(\frac{y^4}{4}\right) = 0$$

yoki

$$d\left[\frac{x^4}{4} + \frac{(xy)^2}{2} + \frac{y^4}{4}\right] = 0$$

ko'rinishda yozish mumkin. Demak,

$$x^4 + 2(xy)^2 + y^4 = C$$

(4) tenglamaning umumiy integrali bo'ladi.

Mustaqil yechish uchun masalalar

To'liq differentsialli tenglamani integrallang.

$$149. x(2x^2 + y^2) + y(x^2 + 2y^2)y' = 0, \quad 150. (3x^2 + 6xy^2)dx + (6x^2y + 4y^3)dy = 0.$$

$$151. \left(\frac{x}{\sqrt{x^2 + y^2}} + \frac{1}{x} + \frac{1}{y}\right)dx + \left(\frac{y}{\sqrt{x^2 + y^2}} + \frac{1}{y} - \frac{x}{y^2}\right)dy = 0.$$

$$152. \left(3x^2 \operatorname{tg} y - \frac{2y^3}{x^3}\right)dx + \left(x^3 \sec^2 y + 4y^3 + \frac{3y^2}{x^2}\right)dy = 0.$$

$$153. \left(2x + \frac{x^2 + y^2}{x^2 y} \right) dx = \left(\frac{x^2 + y^2}{xy^2} \right) dy$$

$$154. \left(\frac{\sin 2x}{y} + x \right) dx + \left(y - \frac{\sin^2 x}{y^2} \right) dy = 0$$

$$155. (3x^2 - 2x - y) dx + (2y - x + 3y^2) dy = 0$$

$$156. \left(\frac{xy}{\sqrt{1+x^2}} + 2xy - \frac{x}{y} \right) dx + (\sqrt{1+x^2} + x^2 - \ln x) dy = 0$$